Ethan Nixon's Portfolio

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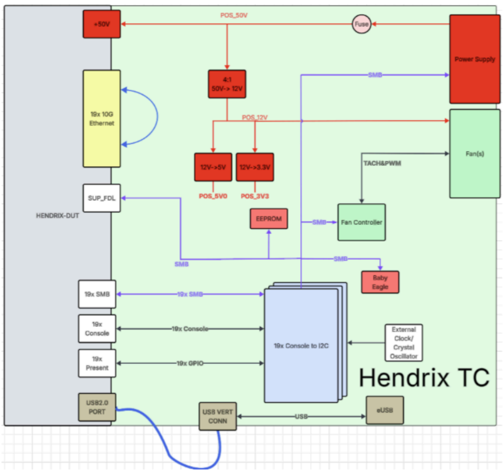
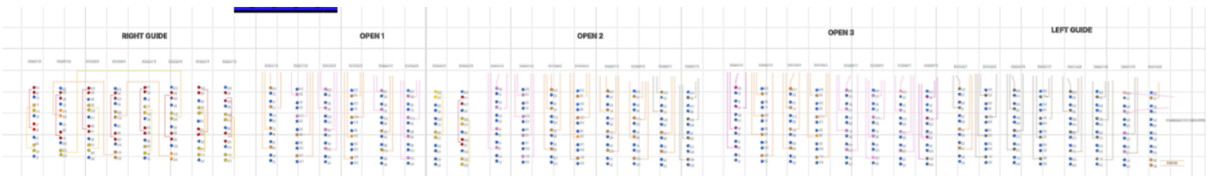
Arista Networks Internship (Hardware Design Intern) Projects+ Validation

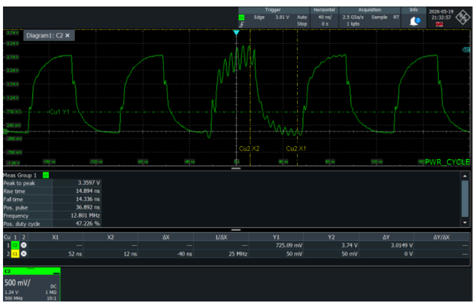
CPU Management Test Card

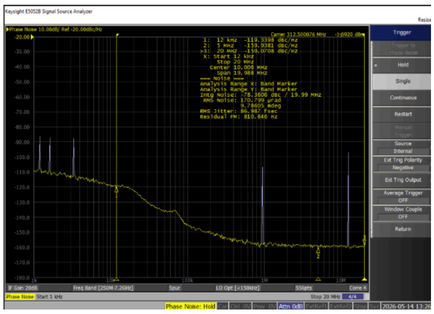
- Designed Schematics, Architecture and Authored Technical document on a RPU (CPU) management Test card able to validate **19x ; 10G Ethernet, SMBus, console links.**

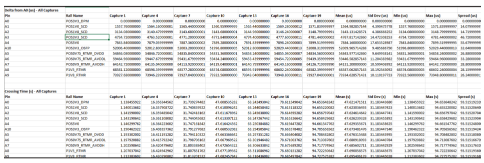


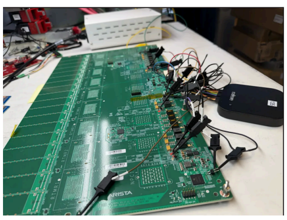


Various Validation & Testing Across different switch cards + test cards.









-^^^ (Bottom Right)-Power Sequence Validation for an 18xOSFP Retimer Card.

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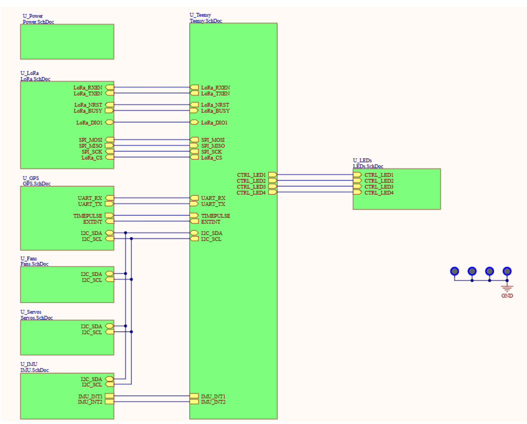
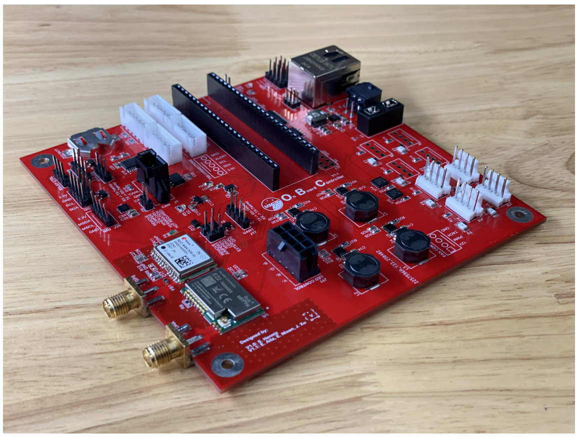
-Automated results with a custom python gui developed.

On Board Controller (OBC)

- Architected and designed a On Board Controller PCB serving as the central communication hub for the rover, using a **Teensy 4.1** to interface with **CAN, I2C, SPI**, and **Ethernet** and communicate with the Jetson AGX Orin
- Integrated **GPS, LoRa**, onboard and external IMU, temperature sensing, fan control, status indicators, and servo motor interfaces into a single embedded control platform
- Designed 12V power distribution with regulated 5V and 3.3V rails and separate RF power domains for GPS and LoRa, supporting up to 6 cooling fans and multiple rover peripherals
- Developed a **4 layer PCB stackup** with dedicated ground and power planes, designed around an 11.8V to 12.2V input range and 10.73A system current requirement
- Defined MCU pin mapping, PWM control, CAN interfaces, I2C addressing, and peripheral communication architecture to support firmware integration across rover subsystems

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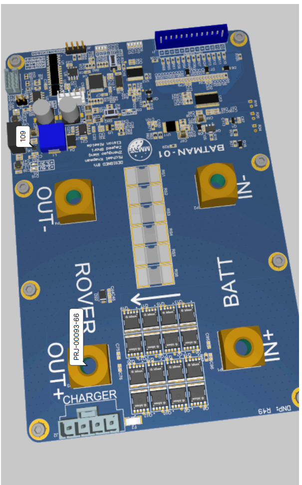
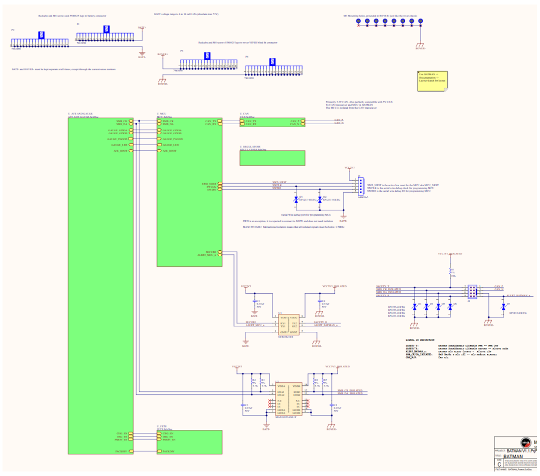
Batman-6S Battery Management System

- Architected and designed a **6S Battery Management System** for a competition rover, integrating cell voltage, pack current, temperature, state of charge monitoring, passive cell balancing, and CAN communication through an STM32 MCU
- Integrated a **BQ76940 Analog Front End, BQ78350 gas gauge, and BQ76200 MOSFET driver** to implement battery measurement, protection, charge and discharge control, and state of charge estimation across the battery stack
- Designed high current protection architecture supporting up to **60A continuous design current**, using parallel low RDSon MOSFETs and bidirectional charge and discharge isolation for overcurrent, short circuit, overvoltage, undervoltage, and temperature fault conditions.
- Implemented pack current sensing using **15 parallel 2 mΩ sense resistors for an effective 0.13 mΩ resistance**, enabling overcurrent and short circuit detection while balancing measurement resolution against thermal dissipation requirements.

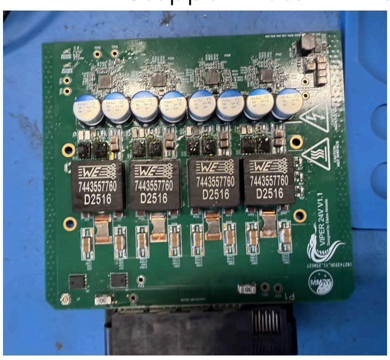
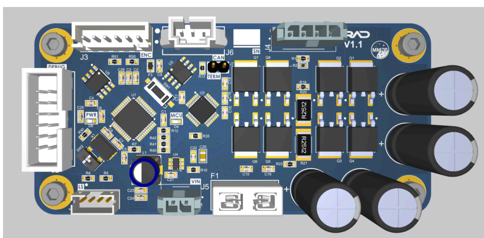




Validation +Testing on 10+ RAD Boards and VIPER 24V card

Viper 24V (left)
- Converts 39.6 V to 50.4 V battery rail into regulated 24 V output
- Uses 4 synchronized MAX15157 buck phases for current sharing

RAD-24V Stepper Motor Driver Board(right)

(right image) RAD-24V Stepper Motor Driver Board

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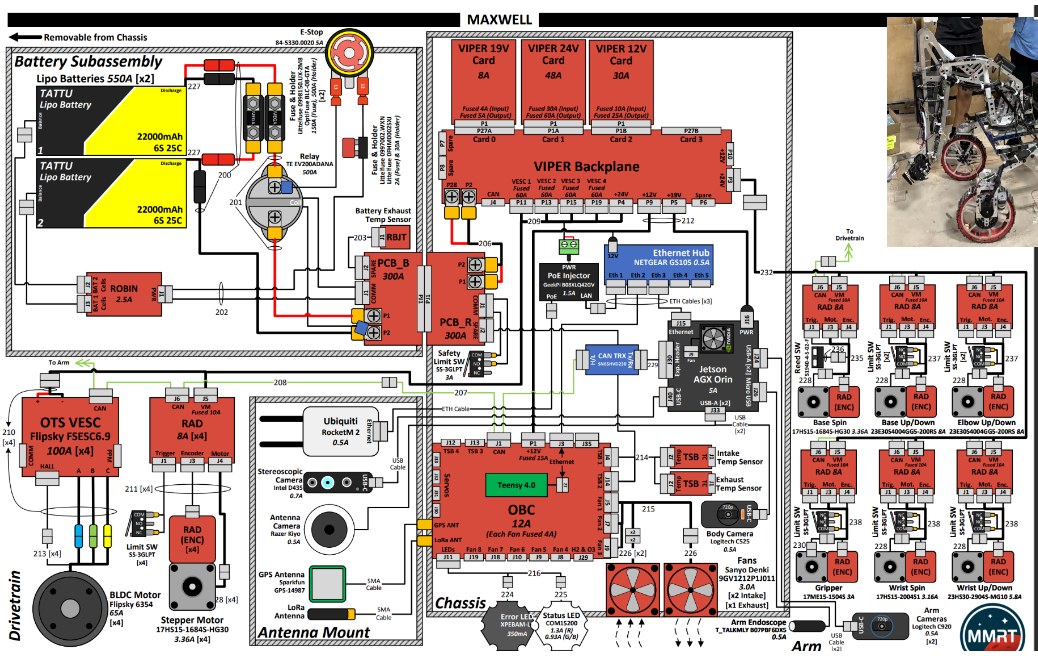
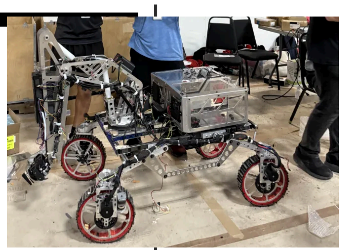
V2.5 and V3 Interconnect





- Designed V2.5 and V3 Electrical Architecture

Electrical Team Lead - [McMaster Mars Rover]()

- Leading & managing 15+ members to create PCBs such as DC-DC converters, RF communications, BMS, and stepper motor controllers, overseeing schematic & PCB reviews, board validation, and assembly.
- Developing a custom 6-layer Battery Management System (BMS) PCB featuring an STM32, passive balancing, charge and discharge isolation, and current/voltage sensing for state of charge calculation.
- Designed and validated a 4-layer Teensy 4.1-based communication board integrating GPS and LoRa modules with dual RF power domains, fan controllers, a servo driver, and diagnostic indicators for telemetry and autonomous control operations.
- Assembled and tested 15+ 24 V stepper motor controller boards, evaluating electrical performance, efficiency, and thermal stability under varying loads.
- Assembled and validated a 6-layer, 4-phase 48 V → 24 V buck converter with integrated fault indicators, soft-start behavior, custom thermal mitigation, and an overvoltage protection stage